

LECTURE 06

LIMIT

- In calculus and its applications, we are often interested in the values $f(x)$ of a function f when x is very close to a number a , but not necessarily equal to a .
- As a matter of fact, in many instances the number a is not in the domain of f ; that is $f(a)$ is undefined.
- Roughly speaking, we ask the following questions:
 - As x gets closer and closer to a (but $x \neq a$), does $f(x)$ get closer and closer to some number L ?
 - If the answer is yes, we say that the limit of $f(x)$, as x approaches a , equals L , and we use the following **limit notation**:

$$\lim_{x \rightarrow a} f(x) = L.$$

EXAMPLE

Evaluate limit numerically as x approaches to 2

$$f(x) = \frac{x^3 - 8}{x - 2}$$

Solution:

L.H.L

x	1.5	1.7	1.9	1.99	1.999
f(x)	9.25	10.29	11.41	11.9401	11.99401

$$\lim_{x \rightarrow 2^-} \frac{x^3 - 8}{x - 2} = 12$$

CONT...

R.H.L

x	2.5	2.3	2.1	2.01	2.001
f(x)	15.25	13.89	12.61	12.0601	12.006

$$\lim_{x \rightarrow 2^+} \frac{x^3 - 8}{x - 2} = 12$$

$$L.H.L = R.H.L$$

$$\Rightarrow \lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = 12$$

EXAMPLE

$$\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x + 3}$$

Solution:

$$\begin{aligned} &= \lim_{x \rightarrow -3} \frac{x^2 + 3x + 2x - 6}{x + 3} \\ &= \lim_{x \rightarrow -3} \frac{(x + 3)(x - 2)}{x + 3} \\ &= \lim_{x \rightarrow -3} (x - 2) \\ &= -3 - 2 \\ &= -5 \end{aligned}$$

L'HÔPITAL'S RULE

- L'Hôpital's rule is a method used to evaluate the limit of a fraction of two functions, $f(x)/g(x)$, when direct substitution results in an indeterminate form like $0/0$ or ∞/∞ .
- The rule states that, under certain conditions, the limit of the quotient $f(x)/g(x)$ is equal to the limit of the quotient of their derivatives, $f'(x)/g'(x)$.
- This process can be repeated if the new limit is still indeterminate, but it only applies to indeterminate forms, not determinate ones.

EXAMPLE

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x}$$

Solution:

After putting direct limit, we have the form 0/0

So applying L'Hôpital's rule, we get

$$\lim_{x \rightarrow 0} \frac{\frac{d}{dx}(\sqrt{x+1} - 1)}{\frac{d}{dx}(x)} = \lim_{x \rightarrow 0} \frac{\frac{1}{2\sqrt{x+1}}}{1} = \frac{1}{2\sqrt{0+1}} = \frac{1}{2}$$

THE LIMIT OF A CONSTANT

➤ If a and c any real numbers, then $\lim_{x \rightarrow a} c = c$.

For example,

$$\lim_{x \rightarrow 3} 8 = 8, \quad \lim_{x \rightarrow 8} 3 = 3, \quad \lim_{x \rightarrow \pi} \sqrt{2} = \sqrt{2}.$$

➤ If a , b and m any real numbers, then $\lim_{x \rightarrow a} (mx + b) = ma + b$.

For example,

$$\lim_{x \rightarrow 4} (3x - 5) = 3(4) - 5 = 12 - 5 = 7,$$
$$\lim_{x \rightarrow \sqrt{2}} (13x + \sqrt{2}) = 13(\sqrt{2}) + \sqrt{2} = 14\sqrt{2}.$$

RULES FOR LIMITS

i. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$

ii. $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$

iii. $\lim_{x \rightarrow a} [f(x)/g(x)] = \lim_{x \rightarrow a} f(x) / \lim_{x \rightarrow a} g(x)$

iv. $\lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$

EXAMPLE

$$\lim_{x \rightarrow 2} \frac{3x + 4}{5x + 7}$$

Solution:

$$\begin{aligned} \frac{\lim_{x \rightarrow 2} (3x + 4)}{\lim_{x \rightarrow 2} (5x + 7)} &= \frac{(3(2) + 4)}{(5(2) + 7)} \\ &= \frac{10}{17} \end{aligned}$$

In Exercises 1–16 use algebraic simplifications to help find the limits, if they exist.

$$1 \quad \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$2 \quad \lim_{x \rightarrow 3} \frac{2x^3 - 6x^2 + x - 3}{x - 3}$$

$$3 \quad \lim_{x \rightarrow 1} \frac{x^2 - x}{2x^2 + 5x - 7}$$

$$4 \quad \lim_{r \rightarrow -3} \frac{r^2 + 2r - 3}{r^2 + 7r + 12}$$

$$5 \quad \lim_{x \rightarrow 5} \frac{3x^2 - 13x - 10}{2x^2 - 7x - 15}$$

$$6 \quad \lim_{x \rightarrow 25} \frac{\sqrt{x} - 5}{x - 25}$$

$$7 \quad \lim_{k \rightarrow 4} \frac{k^2 - 16}{\sqrt{k} - 2}$$

$$8 \quad \lim_{h \rightarrow 0} \frac{(x + h)^3 - x^3}{h}$$

$$9 \quad \lim_{h \rightarrow 0} \frac{(x + h)^2 - x^2}{h}$$

$$10 \quad \lim_{h \rightarrow 2} \frac{h^3 - 8}{h^2 - 4}$$

$$\mathbf{11} \quad \lim_{h \rightarrow -2} \frac{h^3 + 8}{h + 2}$$

$$\mathbf{12} \quad \lim_{z \rightarrow 10} \frac{1}{z - 10}$$

$$\mathbf{13} \quad \lim_{x \rightarrow -3/2} \frac{2x + 3}{4x^2 + 12x + 9}$$

$$\mathbf{14} \quad \lim_{s \rightarrow -1} \frac{1}{s^2 + 2s + 1}$$

$$\mathbf{15} \quad \lim_{x \rightarrow 0} \frac{1}{x^2}$$

$$\mathbf{16} \quad \lim_{t \rightarrow 1} \frac{(1/t) - 1}{t - 1}$$

CONT...

If n is a positive integer, then

i. $\lim_{x \rightarrow a} x^n = a^n.$

ii. $\lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n$, provided $\lim_{x \rightarrow a} f(x)$ exists.

Example:

$$\begin{aligned}\lim_{x \rightarrow 2} (3x + 4)^5 &= \left[\lim_{x \rightarrow 2} (3x + 4) \right]^5 = [3(2) + 4]^5 \\ &= [10]^5 = 100,000.\end{aligned}$$

EXAMPLE

Find $\lim_{x \rightarrow -2} (5x^3 + 3x^2 - 6)$.

Solution:

$$\begin{aligned}\lim_{x \rightarrow -2} (5x^3 + 3x^2 - 6) &= \lim_{x \rightarrow -2} (5x^3) + \lim_{x \rightarrow -2} (3x^2) - \lim_{x \rightarrow -2} (6) \\ &= 5 \lim_{x \rightarrow -2} (x^3) + 3 \lim_{x \rightarrow -2} (x^2) - 6 \\ &= 5((-2)^3) + 3((-2)^2) - 6 \\ &= 5(-8) + 3(4) - 6 \\ &= -34\end{aligned}$$

THEOREM

If $a > 0$ and n is a positive integer, or if $a \leq 0$ and n is an odd positive integer, then

$$\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}.$$

Example 6: Find $\lim_{x \rightarrow 8} \frac{x^{2/3} + 3\sqrt{x}}{4 - (16/x)}.$

Solution:

$$\frac{\lim_{x \rightarrow 8} (x^{2/3} + 3\sqrt{x})}{\lim_{x \rightarrow 8} (4 - (16/x))} = \frac{\lim_{x \rightarrow 8} x^{2/3} + \lim_{x \rightarrow 8} 3\sqrt{x}}{\lim_{x \rightarrow 8} 4 - \lim_{x \rightarrow 8} (16/x)}$$

CONT...

$$\begin{aligned} &= \frac{(8^{2/3} + 3\sqrt{8})}{(4 - (16/8))} \\ &= \frac{(8^{2/3} + 3\sqrt{8})}{(4 - (16/8))} \\ &= \frac{(4 + 6\sqrt{2})}{(4 - 2)} \\ &= \frac{4 + 6\sqrt{2}}{2} \\ &= \frac{4}{2} + \frac{6\sqrt{2}}{2} = 2 + 3\sqrt{2} \end{aligned}$$

THEOREM

If a function f has a limit as x approaches a , then

$$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$$

Provided either n is an odd positive integer or n is an even positive integer and $\lim_{x \rightarrow a} f(x) > 0$.

EXAMPLE 7

Find $\lim_{x \rightarrow 5} \sqrt[3]{3x^2 - 4x + 9}$.

Solution:

$$\begin{aligned}\lim_{x \rightarrow 5} \sqrt[3]{3x^2 - 4x + 9} &= \sqrt[3]{\lim_{x \rightarrow 5} (3x^2 - 4x + 9)} \\ &= \sqrt[3]{(3(5)^2 - 4(5) + 9)} \\ &= \sqrt[3]{75 - 20 + 9} \\ &= \sqrt[3]{64} = 4\end{aligned}$$

LIMIT INVOLVING INFINITY

$$\lim_{x \rightarrow 1} \frac{1}{x - 1}$$

Solution:

L.H.L

x	0.5	0.7	0.9	0.99	0.999
f(x)	-2	-3.33	-10	-100	-1000

$$\lim_{x \rightarrow 1^-} \frac{1}{x - 1} = -\infty$$

CONT...

R.H.L

x	1.5	1.3	1.1	1.01	1.001
f(x)	2	3.33	10	100	1000

$$\lim_{x \rightarrow 1^+} \frac{1}{x-1} = \infty$$

$$L.H.L \neq R.H.L$$

$\Rightarrow$ Limit does not exist

EXAMPLE

$$\lim_{x \rightarrow \pm\infty} \frac{2x + 1}{x + 1}$$

Solution:

L.H.L

x	1	10	100	1000	10000
f(x)	1.5	1.9	1.99	1.999	1.9999

$$\lim_{x \rightarrow +\infty} \frac{2x + 1}{x + 1} = \lim_{x \rightarrow +\infty} \frac{2 + 1/x}{1 + 1/x} = \frac{2 + 1/\infty}{1 + 1/\infty} = \frac{2 + 0}{1 + 0} = 2$$

CONT...

R.H.L

x	-10	-100	-1000	-10000
f(x)	2.1	2.01	2.001	2.0001

$$\lim_{x \rightarrow -\infty} \frac{2x + 1}{x + 1} = \lim_{x \rightarrow -\infty} \frac{2 + 1/x}{1 + 1/x} = \frac{2 + 1/\infty}{1 + 1/\infty} = \frac{2 + 0}{1 + 0} = 2$$

$$L.H.L = R.H.L$$

$\Rightarrow$ Limit exists

In Exercises 1–36 find the limits, if they exist.

$$1 \quad \lim_{x \rightarrow -2} (3x^3 - 2x + 7)$$

$$2 \quad \lim_{x \rightarrow 4} (5x^2 - 9x - 8)$$

$$3 \quad \lim_{x \rightarrow \sqrt{2}} (x^2 + 3)(x - 4)$$

$$4 \quad \lim_{t \rightarrow -3} (3t + 4)(7t - 9)$$

$$5 \quad \lim_{x \rightarrow 4} \sqrt[3]{x^2 - 5x - 4}$$

$$6 \quad \lim_{x \rightarrow -2} \sqrt{x^4 - 4x + 1}$$

$$7 \quad \lim_{x \rightarrow 7} 0$$

$$8 \quad \lim_{x \rightarrow 1/2} \frac{4x^2 - 6x + 3}{16x^3 + 8x - 7}$$

$$9 \quad \lim_{x \rightarrow \sqrt{2}} 15$$

$$10 \quad \lim_{x \rightarrow 15} \sqrt{2}$$

$$11 \quad \lim_{x \rightarrow 1/2} \frac{2x^2 + 5x - 3}{6x^2 - 7x + 2}$$

$$13 \quad \lim_{x \rightarrow 2} \frac{x - 2}{x^3 - 8}$$

$$15 \quad \lim_{x \rightarrow 16} \frac{x - 16}{\sqrt{x} - 4}$$

$$17 \quad \lim_{s \rightarrow 4} \frac{6s - 1}{2s - 9}$$

$$19 \quad \lim_{x \rightarrow 1} \left(\frac{x^2}{x - 1} - \frac{1}{x - 1} \right)$$

$$12 \quad \lim_{x \rightarrow -3} \frac{x + 3}{(1/x) + (1/3)}$$

$$14 \quad \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{(x - 2)^2}$$

$$16 \quad \lim_{x \rightarrow -2} \frac{x^3 + 8}{x^4 - 16}$$

$$18 \quad \lim_{x \rightarrow \pi} (x - 3.1416)$$

$$20 \quad \lim_{x \rightarrow 1} \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^6$$

$$21 \quad \lim_{x \rightarrow 16} \frac{2\sqrt{x} + x^{3/2}}{\sqrt[4]{x} + 5}$$

$$22 \quad \lim_{x \rightarrow -8} \frac{16x^{2/3}}{4 - x^{4/3}}$$

$$23 \quad \lim_{x \rightarrow 3} \sqrt[3]{\frac{2 + 5x - 3x^3}{x^2 - 1}}$$

$$24 \quad \lim_{x \rightarrow \pi} \sqrt[5]{\frac{x - \pi}{x + \pi}}$$

$$25 \quad \lim_{h \rightarrow 0} \frac{4 - \sqrt{16 + h}}{h}$$

$$26 \quad \lim_{h \rightarrow 0} \left(\frac{1}{h} \right) \left(\frac{1}{\sqrt{1 + h}} - 1 \right)$$

$$27 \quad \lim_{x \rightarrow 1} \frac{(x - 1)^5}{x^5 - 1}$$

$$28 \quad \lim_{x \rightarrow 6} (x + 4)^3(x - 6)^2$$

$$29 \quad \lim_{v \rightarrow 3} v^2(3v - 4)(9 - v^3)$$

$$30 \quad \lim_{k \rightarrow 2} \sqrt{3k^2 + 4} \sqrt[3]{3k + 2}$$

$$31 \quad \lim_{t \rightarrow -1} \frac{(4t^2 + 5t - 3)^3}{(6t + 5)^4}$$

$$32 \quad \lim_{t \rightarrow 7} \frac{\sqrt[5]{3 - 5t}}{(t - 5)^3}$$

$$33 \quad \lim_{x \rightarrow 9} \frac{x^2 - 81}{3 - \sqrt{x}}$$

$$34 \quad \lim_{x \rightarrow 8} \frac{x - 8}{\sqrt[3]{x} - 2}$$

$$35 \quad \lim_{h \rightarrow 0} \frac{(2 + h)^{-2} - 2^{-2}}{h}$$

$$36 \quad \lim_{h \rightarrow 0} \frac{(9 + h)^{-1} - 9^{-1}}{h}$$

In Exercises 1–22, find the limits, if they exist.

1 (a) $\lim_{x \rightarrow 5^-} \sqrt{5 - x}$

(b) $\lim_{x \rightarrow 5^+} \sqrt{5 - x}$

(c) $\lim_{x \rightarrow 5} \sqrt{5 - x}$

3 (a) $\lim_{x \rightarrow 1^-} \sqrt[3]{x^3 - 1}$

(b) $\lim_{x \rightarrow 1^+} \sqrt[3]{x^3 - 1}$

(c) $\lim_{x \rightarrow 1} \sqrt[3]{x^3 - 1}$

2 (a) $\lim_{x \rightarrow 2^-} \sqrt{8 - x^3}$

(b) $\lim_{x \rightarrow 2^+} \sqrt{8 - x^3}$

(c) $\lim_{x \rightarrow 2} \sqrt{8 - x^3}$

4 (a) $\lim_{x \rightarrow -8^+} x^{2/3}$

(b) $\lim_{x \rightarrow -8^-} x^{2/3}$

(c) $\lim_{x \rightarrow -8} x^{2/3}$

$$5 \quad (a) \quad \lim_{x \rightarrow 4^-} \frac{|x - 4|}{x - 4}$$

$$(b) \quad \lim_{x \rightarrow 4^+} \frac{|x - 4|}{x - 4}$$

$$(c) \quad \lim_{x \rightarrow 4} \frac{|x - 4|}{x - 4}$$

$$7 \quad \lim_{x \rightarrow 0^+} (4 + \sqrt{x})$$

$$9 \quad \lim_{x \rightarrow -6^+} (\sqrt{x + 6} + x)$$

$$11 \quad \lim_{x \rightarrow 5^+} (\sqrt{x^2 - 25} + 3)$$

$$6 \quad (a) \quad \lim_{x \rightarrow -5^+} \frac{x + 5}{|x + 5|}$$

$$(b) \quad \lim_{x \rightarrow -5^-} \frac{x + 5}{|x + 5|}$$

$$(c) \quad \lim_{x \rightarrow -5} \frac{x + 5}{|x + 5|}$$

$$8 \quad \lim_{x \rightarrow 0^+} (4x^{3/2} - \sqrt{x} + 3)$$

$$10 \quad \lim_{x \rightarrow 5/2^-} (\sqrt{5 - 2x} - x^2)$$

$$12 \quad \lim_{x \rightarrow 3^-} x\sqrt{9 - x^2}$$

$$13 \quad \lim_{x \rightarrow 2^+} \frac{4 - x^2}{2 - x}$$

$$15 \quad \lim_{x \rightarrow 3^+} \frac{\sqrt{(x - 3)^2}}{x - 3}$$

$$17 \quad \lim_{x \rightarrow 5^+} \frac{1 + \sqrt{2x - 10}}{x + 3}$$

$$19 \quad \lim_{x \rightarrow -7^+} \frac{x + 7}{|x + 7|}$$

$$21 \quad \lim_{x \rightarrow 0^+} \frac{1}{x}$$

$$14 \quad \lim_{x \rightarrow -4^+} \frac{2x^2 + 5x - 12}{x^2 + 3x - 4}$$

$$16 \quad \lim_{x \rightarrow -10^-} \frac{x + 10}{\sqrt{(x + 10)^2}}$$

$$18 \quad \lim_{x \rightarrow 4^+} \frac{\sqrt[4]{x^2 - 16}}{x + 4}$$

$$20 \quad \lim_{x \rightarrow \pi^-} \frac{|\pi - x|}{x - \pi}$$

$$22 \quad \lim_{x \rightarrow 8^-} \frac{1}{x - 8}$$